ADVANCED OPTIMISATION

Information for candidates:

- Here you will get reminders of complicated mnemonic expressions if needed in the exam.
- An example of what you may get in this page is the composition theorem for the preservation of convexity and concavity, because that theorem is long and difficult to remember.
- You will NOT get reminders of the fundamental formulas of convexity. For instance you need to know the definitions of convex set, convex function and so on. You will NOT be asked to state these definitions (so you do not have to worry about tricky details), but you will be asked to USE them in exercises; for instance to prove that a function is convex.
- As an example, this mock paper would not have any reminder because no advanced formulas are required. Hence, if you use a formula in this mock paper, it means that you need to know it.

Note that this mock exam has not been checked by the second marker and external examiner, and so it may contain typos. The real exam has been checked by the second marker and will be reviewed by the external examiner, so the chance of having a typo is very remote. If you think there is a typo in the real exam, ask the invigilator.

1. a) Consider a system in which a central node transmits messages to n receivers. Each receiver channel $i \in \{1,\dots,n\}$ has a transmit power P_i and bandwidth W_i. The sums of the transmit power P_i and bandwidth W_i of all channels is P_{tot} and W_{tot}, respectively. Assuming that each channel is corrupted by Gaussian white noise results in a bit rate R_i of each channel given by

$$R_i = \alpha_i W_i \log\left(1+\beta_i \frac{P_i}{W_i}\right),$$

where α_i and β_i are known positive constants.

Given some utility function of the bit rate of each channel, $u_i(R_i)$, the objective is to maximise the total utility

$$U = \sum_{i=1}^{n} u_i(R_i).$$

One of the simplest utility functions is the bit rate itself, which also gives a convex objective function.

i) Formulate the considered problem as a convex minimisation problem.

ii) Although the problem above is convex, it cannot be solved by CVXPY because the software does not recognise the division of two variables as a convex preserving operation. A function which is recognised as convex by CVXPY is the Kullback-Leibler divergence, which is defined as $\text{kl_div}(x,y) = x\log\frac{x}{y} - x + y$. Rewrite the objective function of the problem using the Kullback-Leibler divergence $\text{kl_div}(x,y)$ to eliminate the explicit division of the two variables.

b) The Euclidean distance between the polyhedra $\mathscr{P}_1 = \{x : A_1 x \preccurlyeq b_1\}$ and $\mathscr{P}_2 = \{x : A_2 x \preccurlyeq b_2\}$ is defined as $\textbf{dist}(\mathscr{P}_1,\mathscr{P}_2) = \inf\{||x_1 - x_2||_2 : x_1 \in \mathscr{P}_1, x_2 \in \mathscr{P}_2\}$.

i) Formulate the problem of finding the closest pair of points, one in $\mathscr{P}_1$ and one in $\mathscr{P}_2$, as a quadratic programme.

ii) Describe the geometric characteristics of the two polyhedra when the problem is infeasible;

iii) Describe the geometric characteristics of the two polyhedra when the problem has zero as optimal value.

2. Consider the optimisation problem

$$\begin{aligned} \min \quad & x^2 + 1 \\ s.t. \quad & (x-2)(x-4) \leq 0, \end{aligned}$$

with variable $x \in \mathbb{R}$.

a) Determine the feasible set, the optimal value and the optimal solution.

b) Plot the objective function and the constraint function versus x (in the same plot). In the plot, clearly indicate the feasible set, the optimal point and the optimal value.

c) Determine the Lagrange dual function g.
Hint: there are two cases to consider depending on the value of λ.

d) State the dual problem. Find the dual optimal value and the dual optimal solution λ^*. Sketch the Lagrange dual function g and indicate its maximum value on the plot. Does strong duality hold?

3. a) Prove the following facts.

i) A slab, i.e. a set of the form $\{x \in \mathbb{R}^n : \alpha \leq a^\top x \leq \beta\}$, is convex.

ii) The norm cone, i.e. the set $C = \{(x,t) : ||x|| \leq t\} \subseteq \mathbb{R}^{n+1}$, is convex. Name any inequality that you use.

b) Prove whether the following functions are convex, concave or neither.

i) The max function $f(x_1,\ldots,x_n) = \max\{x_1,\ldots,x_n\}$ on $\mathbb{R}^n$.

ii) $f(x_1,x_2) = \frac{1}{x_1 x_2}$ on $\mathbb{R}^2_{++}$.

c) Solve the squared ℓ_2 least-norm problem.

$$\begin{aligned} \min_x \quad & ||x||_2^2, \\ \text{s.t.} \quad & Ax = b. \end{aligned}$$

d) Formulate the problem

$$\min_x \quad ||Ax - b||_\infty, \tag{3.1}$$

where $A \in \mathbb{R}^{m\times n}$, $b \in \mathbb{R}^m$, and $x \in \mathbb{R}^n$, as a linear programme. Explain in detail the relation between problem (3.1) and the obtained linear programme.

4. Consider the optimisation problem (without equality constraints)

$$\begin{aligned} \min_x \quad & f_0(x) \\ \text{s.t.} \quad & f_i(x) \leq 0, \qquad i = 1, \ldots, m, \end{aligned}$$

where f_i are twice differentiable. We define the log barrier for this problem as

$$\phi(x) = \sum_{i=1}^{m} -\log(-f_i(x)),$$

with domain $\{x : f_i(x) < 0, i = 1, \ldots, m\}$. We define the central path as

$$x^*(t) = \operatorname{argmin} \{t f_0(x) + \phi(x)\},$$

where $t > 0$ is a parameter (we assume that for each t, the minimiser exists and is unique).

a) Let $z = x^*(t)$ be the minimiser of $t f_0(x) + \phi(x)$. Find the dual feasible $\lambda^*(t)$ as function of z. Find the associated duality gap.

b) We can interpret the central path conditions as a continuous deformation of the KKT complementary condition. Using $\lambda^*(t)$ found in part b), determine the deformed complementary condition. Show that as $t \to \infty$, the original complementary condition is recovered.

c) Let $\frac{dx^*(t)}{dt}$ be the tangent to the central path at the point $x^*(t)$. Find an explicit expression for $\frac{dx^*(t)}{dt}$.
Hint: Differentiate the centrality equation (found in part b)) with respect to t.